

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Industry based problem Solving

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4 –Industry based problem Solving	Assessment:	Assessment Tool 2 – Industry based problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Analyze real-world business data using confidence intervals, hypothesis testing, test statistics, and p-value approaches to evaluate claims and support data-driven industry decisions.		
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Section / Batch:	CSE – AI	Date:	20/08/26

Code of Conduct

I Y. Jayesh Ranjan - 192472201 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Y. Jayesh Ranjan

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	

Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	
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Questions:

Question 1 – Digital Marketing

Before introducing a new advertisement, the conversion rate was 8%. After the campaign, 90 out of 1,000 visitors converted. Using a 95% confidence interval, determine whether the observed conversion rate provides evidence of improvement.

Question 2 – Education Technology

An EdTech company claims that its adaptive-learning platform improves the average test score to at least 75. A sample of 64 students obtains an average score of 78 with standard deviation 10. Use a 95% confidence interval to assess the claim.

Question 3 – Automobile Industry

A car manufacturer claims that its new model provides an average mileage of 20 km/litre. A sample of 50 cars gives a mean mileage of 19.2 km/litre with standard deviation 2.5 km/litre. Perform a hypothesis test at the 5% significance level using the p-value approach.

Question 4 – Cloud Computing

A cloud provider claims that its average server response time is 100 ms. A sample of 81 requests has a mean of 104 ms and standard deviation of 12 ms. Test the claim at $\alpha = 0.05$ using the p-value method.

Question 5 – E-commerce

An online company claims that its product return rate is 10%. In a sample of 1,000 orders, 130 products were returned. Test whether the actual return rate is significantly higher than 10% using the p-value approach at the 5% significance level.

PROBLEM 1 – DIGITAL MARKETING

Question: Before the new advertisement, the conversion rate was 8%. After the campaign, 90 out of 1,000 visitors converted. Using a 95% confidence interval, determine whether the observed conversion rate provides evidence of improvement.

1. Identify the population and sample

Population: All visitors who could be exposed to the new advertising campaign and whose conversion behavior is of interest.

Sample: The 1,000 visitors observed after the campaign.

Observed conversions: 90 visitors.

2. Identify the parameter of interest

Parameter: Let p = the true conversion rate of all visitors after the new campaign.

3. Calculate the point estimate

Sample proportion: The point estimate of p is the sample conversion rate, $\hat{p}$.

$$\hat{p} = x / n$$

$$\hat{p} = 90 / 1,000$$

$$\hat{p} = 0.09 = 9\%$$

Therefore, the estimated conversion rate after the campaign is 9%. The observed increase is $9\% - 8\% = 1$ percentage point.

4. State the null hypothesis H_0

$$H_0: p = 0.08$$

The null hypothesis represents the original conversion rate of 8%.

5. State the alternative hypothesis H_1

$$H_1: p > 0.08$$

This is a right-tailed test because an improvement means the new conversion rate should be greater than 8%.

6. Select the significance level α

$$\text{Confidence level} = 95\%$$

$$\alpha = 1 - 0.95 = 0.05$$

Thus, $\alpha = 0.05$.

7. Calculate the 95% confidence interval and test statistic

Because this is a population proportion problem, the 95% confidence interval is:

$$95\% \text{ CI} = \hat{p} \pm z_{0.025} \sqrt{[\hat{p}(1 - \hat{p}) / n]}$$

$$\text{For 95\% confidence, } z_{0.025} = 1.96$$

$$\text{Standard error} = \sqrt{[(0.09)(0.91) / 1,000]}$$

$$\text{Standard error} = \sqrt{(0.0000819)}$$

$$\text{Standard error} \approx 0.00905$$

$$\text{Margin of error} = 1.96(0.00905) \approx 0.01774$$

$$95\% \text{ CI} = 0.09 \pm 0.01774$$

$$95\% \text{ CI} = (0.07226, 0.10774)$$

$$95\% \text{ CI} \approx (7.23\%, 10.77\%)$$

The original 8% conversion rate lies inside this confidence interval.

For the corresponding one-proportion hypothesis test:

$$z = (\hat{p} - p_0) / \sqrt{[p_0(1 - p_0) / n]}$$

$$z = (0.09 - 0.08) / \sqrt{[(0.08)(0.92) / 1,000]}$$

$$z \approx 1.166$$

8. Calculate the p-value

Because $H_1: p > 0.08$, this is a right-tailed test.

$$\text{p-value} = P(Z > 1.166)$$

$$\text{p-value} \approx 0.1219$$

9. Compare the p-value with α

$$\text{p-value} = 0.1219$$

$$\alpha = 0.05$$

Since $0.1219 > 0.05$, we FAIL TO REJECT H_0 .

The result is not statistically significant at the 5% level.

10. Make an industry-oriented conclusion

Industry-oriented conclusion: The sample conversion rate increased from 8% to 9%, but the increase is not statistically significant at the 5% level. The 95% confidence interval is 7.23% to 10.77%, which includes

the original 8% rate. Therefore, the digital marketing campaign does not provide sufficient statistical evidence that the true conversion rate improved. The company should collect more campaign data or run a larger controlled experiment before concluding that the new advertisement improves conversions.

PROBLEM 2 – EDUCATION TECHNOLOGY

Question: An EdTech company claims that its adaptive-learning platform improves the average test score to at least 75. A sample of 64 students has an average score of 78 with standard deviation 10. Use a 95% confidence interval to assess the claim.

1. Identify the population and sample

Population: All students whose test performance is represented by the study.

Sample: The 64 students who used the adaptive-learning platform.

Sample statistics: $\bar{x} = 78$ and $s = 10$.

2. Identify the parameter of interest

Parameter: Let μ = the true population mean test score when using the adaptive-learning platform.

3. Calculate the point estimate

Point estimate of $\mu = \bar{x}$

$$\bar{x} = 78$$

Therefore, the estimated average test score is 78 points.

4. State the null hypothesis H_0

$$H_0: \mu \geq 75$$

The company's claim is that the average score is at least 75. The boundary value 75 is used for the hypothesis calculation.

5. State the alternative hypothesis H_1

$$H_1: \mu < 75$$

This alternative represents evidence against the company's claim: the true average score would be below 75.

6. Select the significance level α

Confidence level = 95%

$$\alpha = 1 - 0.95 = 0.05$$

Thus, $\alpha = 0.05$.

7. Calculate the 95% confidence interval and test statistic

The population standard deviation is unknown, so a one-sample t procedure is appropriate. Degrees of freedom:

$$df = n - 1 = 64 - 1 = 63$$

$$\text{Standard error} = s / \sqrt{n} = 10 / \sqrt{64} = 10 / 8 = 1.25$$

$$\text{For 95\% confidence and } df = 63, t_{0.025, 63} \approx 1.998$$

$$\text{Margin of error} = 1.998(1.25) \approx 2.50$$

$$95\% \text{ CI} = \bar{x} \pm t^* (s / \sqrt{n})$$

$$95\% \text{ CI} = 78 \pm 2.50$$

$$95\% \text{ CI} \approx (75.50, 80.50)$$

Because the entire confidence interval is above 75, the sample provides support for the company's claim that the mean score is at least 75.

For the corresponding test statistic at the boundary $\mu_0 = 75$:

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

$$t = (78 - 75) / (10 / \sqrt{64})$$

$$t = 3 / 1.25$$

$$t = 2.40$$

8. Calculate the p-value

For $H_1: \mu < 75$, the test is left-tailed.

p-value = $P(T_{63} < 2.40)$ is not the appropriate tail because the observed statistic is positive.

For the actual evidence against H_0 , the left-tail probability is very large: p-value ≈ 0.9903 .

Equivalently, the evidence is strongly in the direction of $\mu > 75$, not $\mu < 75$.

Thus, p-value ≈ 0.9903 for the claim-testing direction $H_1: \mu < 75$.

9. Compare the p-value with α

$$\text{p-value} \approx 0.9903$$

$$\alpha = 0.05$$

Since $0.9903 > 0.05$, we FAIL TO REJECT H_0 .

The data do not provide evidence that the true mean is below 75. In fact, the sample mean of 78 and the 95% CI above 75 support the company's claim.

10. Make an industry-oriented conclusion

Industry-oriented conclusion: The adaptive-learning platform meets the company's stated benchmark based on this sample. The estimated mean score is 78, and the 95% confidence interval is approximately 75.50 to 80.50, entirely above the 75-point threshold. Therefore, there is no statistical evidence that the platform fails to achieve an average score of at least 75. The EdTech company can reasonably state that the study supports its 75-point benchmark, while recognizing that additional studies would strengthen the business claim.

PROBLEM 3 – AUTOMOBILE INDUSTRY

Question: A car manufacturer claims that its new model provides an average mileage of 20 km/litre. A sample of 50 cars gives a mean of 19.2 km/litre with standard deviation 2.5 km/litre. Perform a hypothesis test at the 5% significance level using the p-value approach.

1. Identify the population and sample

Population: All cars of the new model produced by the manufacturer.

Sample: The 50 cars tested.

Sample statistics: $\bar{x} = 19.2$ km/litre and $s = 2.5$ km/litre.

2. Identify the parameter of interest

Parameter: Let μ = the true mean mileage of the new model, in km/litre.

3. Calculate the point estimate

Point estimate of $\mu = \bar{x}$

$$\bar{x} = 19.2 \text{ km/litre}$$

4. State the null hypothesis H_0

$$H_0: \mu = 20 \text{ km/litre}$$

5. State the alternative hypothesis H_1

$$H_1: \mu \neq 20 \text{ km/litre}$$

This is a two-tailed test because the question asks whether the actual average differs from 20 km/litre in either direction.

6. Select the significance level α

$$\alpha = 0.05 = 5\%$$

7. Calculate the test statistic

The population standard deviation is unknown, so use a one-sample t-test.

$$df = n - 1 = 50 - 1 = 49$$

$$\text{Standard error} = s / \sqrt{n} = 2.5 / \sqrt{50}$$

$$\sqrt{50} \approx 7.071$$

$$\text{Standard error} \approx 2.5 / 7.071 = 0.3536$$

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

$$t = (19.2 - 20) / 0.3536$$

$$t = -0.8 / 0.3536$$

$$t \approx -2.263$$

Optional confidence-interval check:

$$95\% \text{ CI} = 19.2 \pm t_{0.025, 49}(0.3536)$$

$$t_{0.025, 49} \approx 2.010$$

$$95\% \text{ CI} \approx 19.2 \pm 0.711$$

$$95\% \text{ CI} \approx (18.49, 19.91) \text{ km/litre}$$

8. Calculate the p-value

Because $H_1: \mu \neq 20$, this is a two-tailed test.

$$\text{p-value} = 2P(T_{49} < -2.263)$$

$$\text{p-value} \approx 0.0281$$

9. Compare the p-value with α

$$\text{p-value} = 0.0281$$

$$\alpha = 0.05$$

Since $0.0281 < 0.05$, we REJECT H_0 .

10. Make an industry-oriented conclusion

Industry-oriented conclusion: There is statistically significant evidence that the true average mileage differs from the manufacturer's claimed 20 km/litre. Since the sample mean is only 19.2 km/litre and the 95% confidence interval is approximately 18.49 to 19.91 km/litre, the evidence specifically suggests that the actual average mileage is lower than 20 km/litre. The manufacturer should review the mileage claim and conduct additional testing before using 20 km/litre as a performance benchmark in marketing.

PROBLEM 4 – CLOUD COMPUTING

Question: A cloud provider claims that its average server response time is 100 ms. A sample of 81 requests has a mean of 104 ms and standard deviation of 12 ms. Test the claim at $\alpha = 0.05$ using the p-value method.

1. Identify the population and sample

Population: All server requests handled by the cloud provider under the conditions represented by the study.

Sample: The 81 requests observed.

Sample statistics: $\bar{x} = 104$ ms and $s = 12$ ms.

2. Identify the parameter of interest

Parameter: Let μ = the true mean server response time, in milliseconds.

3. Calculate the point estimate

Point estimate of $\mu = \bar{x}$

$$\bar{x} = 104 \text{ ms}$$

4. State the null hypothesis H_0

$$H_0: \mu = 100 \text{ ms}$$

5. State the alternative hypothesis H_1

$$H_1: \mu \neq 100 \text{ ms}$$

This is a two-tailed test because the claim is being tested for any difference from 100 ms.

6. Select the significance level α

$$\alpha = 0.05 = 5\%$$

7. Calculate the test statistic

The population standard deviation is unknown, so use a one-sample t-test.

$$df = n - 1 = 81 - 1 = 80$$

$$\text{Standard error} = s / \sqrt{n} = 12 / \sqrt{81}$$

$$\text{Standard error} = 12 / 9 = 1.3333 \text{ ms}$$

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

$$t = (104 - 100) / 1.3333$$

$$t = 4 / 1.3333$$

$$t = 3.00$$

Optional confidence-interval check:

$$95\% \text{ CI} = 104 \pm t_{0.025,80}(1.3333)$$

$$t_{0.025,80} \approx 1.990$$

$$95\% \text{ CI} \approx 104 \pm 2.653$$

$$95\% \text{ CI} \approx (101.35, 106.65) \text{ ms}$$

8. Calculate the p-value

Because $H_1: \mu \neq 100$, this is a two-tailed test.

$$\text{p-value} = 2P(T_{80} > 3.00)$$

$$\text{p-value} \approx 0.0036$$

9. Compare the p-value with α

$$\text{p-value} = 0.0036$$

$$\alpha = 0.05$$

Since $0.0036 < 0.05$, we REJECT H_0 .

10. Make an industry-oriented conclusion

Industry-oriented conclusion: There is statistically significant evidence that the true average server response time differs from the claimed 100 ms. The sample mean is 104 ms, and the 95% confidence interval is approximately 101.35 to 106.65 ms, entirely above 100 ms. Therefore, the evidence indicates that the cloud provider's actual average response time is higher than 100 ms. The provider should investigate latency and system performance before continuing to present 100 ms as its average response-time claim.

PROBLEM 5 – E-COMMERCE

Question: An online company claims that its product return rate is 10%. In a sample of 1,000 orders, 130 products were returned. Test whether the actual return rate is significantly higher than 10% using the p-value approach at the 5% significance level.

1. Identify the population and sample

Population: All orders/products sold by the online company for which the return rate is being evaluated.

Sample: The 1,000 orders observed.

Returned products: 130 products were returned.

2. Identify the parameter of interest

Parameter: Let p = the true population return rate.

3. Calculate the point estimate

$$\hat{p} = x / n$$

$$\hat{p} = 130 / 1,000$$

$$\hat{p} = 0.13 = 13\%$$

The estimated actual return rate is therefore 13%.

4. State the null hypothesis H_0

$$H_0: p = 0.10$$

5. State the alternative hypothesis H_1

$$H_1: p > 0.10$$

This is a right-tailed test because the question specifically asks whether the actual return rate is higher than 10%.

6. Select the significance level α

$$\alpha = 0.05 = 5\%$$

7. Calculate the test statistic

For a one-sample proportion test, use the null value $p_0 = 0.10$ when calculating the standard error.

$$\text{Standard error under } H_0 = \sqrt{[p_0(1 - p_0) / n]}$$

$$SE = \sqrt{[(0.10)(0.90) / 1,000]}$$

$$SE = \sqrt{(0.00009)}$$

$$SE \approx 0.00949$$

$$z = (\hat{p} - p_0) / SE$$

$$z = (0.13 - 0.10) / 0.00949$$

$$z = 0.03 / 0.00949$$

$$z \approx 3.162$$

Optional 95% confidence-interval check:

$$95\% \text{ CI} = \hat{p} \pm 1.96\sqrt{[\hat{p}(1 - \hat{p})/n]}$$

$$SE = \sqrt{[(0.13)(0.87)/1,000]} \approx 0.01064$$

$$\text{Margin of error} \approx 1.96(0.01064) \approx 0.02085$$

$$95\% \text{ CI} \approx 0.13 \pm 0.02085$$

$$95\% \text{ CI} \approx (0.1092, 0.1508)$$

$$95\% \text{ CI} \approx (10.92\%, 15.08\%)$$

8. Calculate the p-value

Because $H_1: p > 0.10$, this is a right-tailed z-test.

$$\text{p-value} = P(Z > 3.162)$$

$$\text{p-value} \approx 0.00078$$

9. Compare the p-value with α

$$\text{p-value} \approx 0.00078$$

$$\alpha = 0.05$$

Since $0.00078 < 0.05$, we REJECT H_0 .

The result is highly statistically significant.

10. Make an industry-oriented conclusion

Industry-oriented conclusion: There is very strong statistical evidence that the actual product return rate is higher than the company's claimed 10%. The sample return rate is 13%, and the p-value is approximately 0.00078, which is much smaller than 0.05. The 95% confidence interval is approximately 10.92% to 15.08%,

also indicating a return rate above 10%. The company should investigate causes of returns, such as product quality, inaccurate descriptions, sizing or fit issues, customer expectations, or shipping damage, because a higher return rate can increase costs and reduce profitability.